

RiskPulse

Financial risk intelligence
and mock index rebalancing

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Case study scope

Financial news and company announcements feed one engine.

Required output	Representation	Application
Sentiment score	-1.0 to +1.0	Mock index allocation
Event classification	Categorical label	Risk search and filtering
Impact score	1 to 10 severity proxy	High-impact signal view

Selected downstream module: A

Ten stocks, current weights, and persisted allocation history.

System architecture

Component	Implementation	Output
Ingestion	HTTPX and RSS/Atom parsing	Clean source records
Risk engine	FinBERT plus event/severity rules	Structured risk signals
Persistence	SQLite	Signals and snapshots
Rebalancer	Confidence and recency tilt	Constrained weights
API and dashboard	FastAPI and React	JSON, CSV and charts

Live and replay portfolios stay separate.

An inspectable risk signal

Synthetic: JPMorgan faces a major liquidity crisis and missed repayment.

Signal field	Actual output
Sentiment	-0.9585
Event	Credit Event
Impact	10 / 10
Confidence	96.73%

Sentiment

FinBERT estimates the financial tone of the company context.

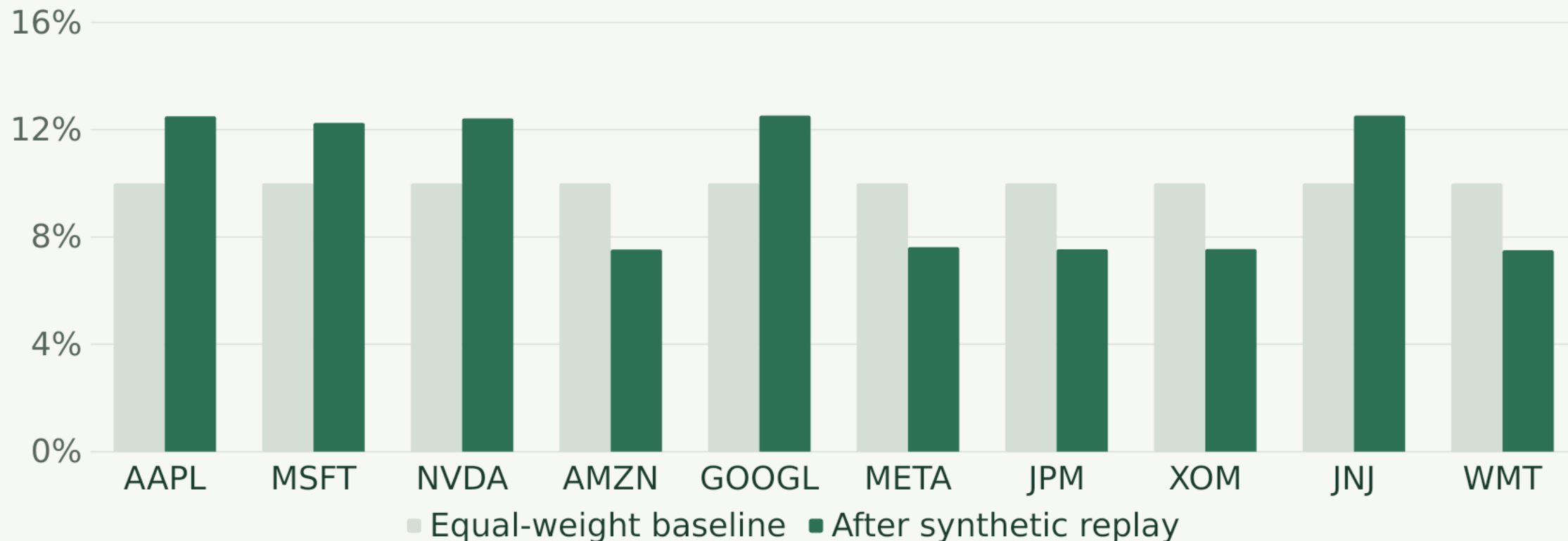
Event and severity

Matched phrases and severity cues explain the rule output.

Impact is a severity estimate. It does not forecast a price change.

Sentiment changes index weights

12 fictional events. FinBERT backend. Separate replay portfolio.



100% allocation. Stock limits: 2% to 18%. Normal turnover limit: 10%.

Validation results

Check	Observed result
Implementation	28 automated tests passed
Live ingestion	50 items across 3 connectors
Synthetic labels	24/24 sentiment, 24/24 event
Analysis time	80.47 ms mean, 95.44 ms p95

Evaluation scope

24 synthetic sanity cases. Loaded CPU model. Network fetch excluded.

Real-news accuracy, market impact and portfolio returns need further validation.

Challenges and next steps

Operational challenges

Repeated provider headlines
Fingerprints suppress duplicates.

Mixed company context
Shared context cannot tilt stocks.

Unavailable feed or model
Visible errors and fallback status.

Further work

Evaluate an independently
labeled real-news dataset.

Calibrate severity and backtest
with transaction costs.

Extend the event API to
Module B stress testing.

AI assistance and third-party sources are disclosed in the repository.